

Crawlspace Conundrum



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Q. I have a problem with the crawlspace in my home.

When the air conditioning is running, I can smell the crawlspace. Is there a good way to keep the crawlspace from smelling? I know I need to seal the ductwork. How do you air seal the floor deck between the living space and the crawlspace? How and where do you install the exhaust fan to depressurize the crawlspace? Is the fan actually blowing into the crawlspace or into the house? Don't you need cross ventilation? If the fan is blowing into the crawlspace, don't you need another vent somewhere to let the air out of it?

—Odoriferous in Ohio

A. Generally, I don't believe that sealing the floor deck is a top priority, especially if the crawlspace is to be partially conditioned. In any event, typical sealing techniques are using caulk and backer rod for small to medium-size gaps and openings, expanding foam for larger openings, and sheet metal or plywood for the largest openings. These openings include electrical, heating/cooling, and

plumbing penetrations; joints in the framing and flooring; and so on. Usually it's easier to access and seal from the crawlspace, but for some gaps it's easier from upstairs.

The air leaks that I'm most concerned about are those in ducts in the

crawlspace and those between the crawlspace and outdoors. Duct leaks can create large pressure imbalances that may draw in pollutants from the soil and circulate crawlspace air to the upstairs. Outdoor air that leaks into the crawlspace can carry with it lots of moisture, and this can lead to mold growth.

Crawlspace air should be exhausted to the outdoors and not into the house. A small, continuously operating exhaust fan, installed to vent crawlspace air to the outdoors, works well. This fan often slightly depressurizes the crawlspace and pulls conditioned air from the upstairs—now you see why I don't emphasize sealing the floor deck, and why sealing the crawlspace from the outside is important. This results in the crawlspace being warmer and drier.

Because your crawlspace smells, I suspect there is a moisture and microbial problem. By eliminating the moisture, you'll probably control the odor. The exhaust fan should help by removing moisture from the space, and by increasing the ventilation with drier upstairs air. You may need to install a poly barrier over the soil floor if moisture migration from the soil is an issue. Be sure to check for plumbing leaks, bad drainage, and poor grading as other causes of moisture problems.



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Odors in the crawlspace can be due to microbial growth caused by moisture accumulation from humid outdoor air or moisture from the soil.

